Jack Lu

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Education

New York University Sep 2023 – Present
Ph.D. in Computer Science (3.98/4.00 GPA)
Advised by: Mengye Ren, thesis committee: Greg Durrett, Kyunghyun Cho
Research focus: Self-Improvement, Reasoning, World Models

University of Waterloo Sep 2018 – May 2023
B.Math. in Computer Science and Mathematics (4.00/4.00 GPA)
Advised by: Raquel Urtasun, Sanja Fidler, Alexander Wong
Research focus: Generative Modeling, Autonomous Driving, ML for Health

Selected Publications

TacitVLA: Reasoning Vision-Language-Action Models Without Reasoning Latency — Under review
Jack Lu, Yurong You, Boyi Li, Yulong Cao, Thomas Tian, Chaowei Xiao, Xiangyu Chen, Wenjie Luo, Boris Ivanovic, Marco Pavone*, Yan Wang*

The Surprising Effectiveness of Deleting Weights in LLM Post-Training — ICMLW FoGen 2026, NeurIPS 2026
Jack Lu*, Zhenbang Yang*, Mike Lasby, Tejas Pote, Yani Ioannou, Mengye Ren

HyperThink: Text-to-Parameter Hypernetworks for Efficient Reasoning — COLM 2026
Donggyun Kim*, **Jack Lu***, Chanwoo Kim, Mengye Ren, Seunghoon Hong

When Does Verification Pay Off? A Closer Look at LLMs as Solution Verifiers — ICLRW RSI 2026
Jack Lu*, Ryan Teehan*, Jinran Jin, Mengye Ren

Context Tuning for In-Context Optimization — ICML 2026
Jack Lu, Ryan Teehan, Zhenbang Yang, Mengye Ren

Solaris: Building a Multiplayer Video World Model in Minecraft — NeurIPS 2026 (Spotlight)
Georgy Savva*, Oscar Michel*, Daohan Lu*, Suppakit Waiwitlikhit, Timothy Meehan, Dhairya Mishra, Srivats Poddar, **Jack Lu**, Saining Xie

SkillFactory: Self-Distillation For Learning Cognitive Behaviors — ICLR 2026
Zayne Sprague, **Jack Lu**, Manya Wadhwa, Sedrick Keh, Mengye Ren, Greg Durrett

ProCreate, Don't Reproduce! Propulsive Energy Diffusion for Creative Generation — ECCV 2024
Jack Lu, Ryan Teehan, Mengye Ren

SceneControl: Diffusion for Controllable Traffic Scene Generation — ICRA 2024
Jack Lu*, Kelvin Wong*, Chris Zhang, Simon Suo, Raquel Urtasun

Research Experience

Visiting Researcher – Meta Superintelligence Labs (FAIR) Sep 2026 – Present
Hosts: Jack Lanchantin, Jason Weston
- Researching recursive self-improvement for large language models.

Research Intern – NVIDIA May 2026 – Sep 2026
Hosts: Yan Wang, Boris Ivanovic, Marco Pavone
- Led TacitVLA: reasoning VLAs that match explicit reasoning at up to 3.4× lower latency.

Ph.D. Student / Research Assistant – CILVR Lab @ NYU Sep 2023 – Present
Advisor: Mengye Ren

- Efficient adaptation: led Context Tuning (prompt optimization) and Deleting Weights (sparse post-training).
- Efficient reasoning: co-led HyperThink, which turns each query into a parameter update instead of a long trace.
- Evaluation and self-improvement: led a 37-model study of LLM verifiers, and co-authored SkillFactory.
- Generative models: led ProCreate (creative diffusion) and contributed to Solaris (multiplayer world model).

Research Intern – Waabi Sep 2022 – Feb 2023
Host: Raquel Urtasun

- Achieved state-of-the-art traffic scene generation performance with diffusion models and graph neural networks (ICRA 2024).

Research Intern – NVIDIA Sep 2021 – Mar 2022
Host: Sanja Fidler

- Applied domain-adversarial training and asset randomization methods to improve monocular 3D object detection networks.

Research Intern – DarwinAI Sep 2020 – Dec 2020
Host: Alexander Wong

- Applied computer vision networks for CT scan analysis to detect terminal lung disease (Frontiers in Artificial Intelligence 2021).

Work Experience

Deep Learning Engineer (Intern) – NVIDIA May 2021 – Aug 2021

- Reduced the failure rate of the autonomous-vehicle path detector by 21% with adversarial training.
- Accelerated collision detection in NVIDIA DriveSim by 7x with a custom KD-Tree algorithm.

Cognitive Software Developer (Intern) – IBM Jan 2020 – Apr 2020

- Improved IBM Cognos Analytics chatbot's named entity recognition performance by 24% with BERT backbone.
- Developed and deployed a novel tabular data clustering algorithm with word embeddings and ontology trees.

Full Stack Developer (Intern) – Deep Trekker May 2019 – Aug 2019

- Engineered a real-time location tracking application that enables remote robot tracking.

Honors and Awards

NSERC – Postgraduate Scholarship, Doctoral (PGS D) ($120,000) 2024 – 2027
University of Waterloo – Winston and Diana Cherry Scholarship ($3,000) 2023
University of Waterloo – President's Research Award ($1,500) 2020
University of Waterloo – President's Scholarship of Distinction ($5,000) 2019
University of Waterloo – Dean's Honours List/Distinction (all terms) 2018 – 2023

Positions of Responsibility

Teaching: Recitation Leader, Deep Learning (NYU, CSCI-GA 2572)
Advising: Zhenbang Yang, Wun Ting Chan, Jinran Jin, Anushka Polapally
Reviewing: CVPR 2024, ICML 2025/2026, ICLR 2025/2026/2027, NeurIPS 2026, and various workshops

Skills

Languages: Python, C++, SQL, LaTeX
Machine Learning: PyTorch, JAX, vLLM, SGLang, verl, DeepSpeed
Others: Slurm, Docker, GCP, Weights & Biases, Git